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SUPERSEDES
MIL-B-83931A (USAF)
05 MAY 1980

# MILITARY SPECIFICATION

# BINDERS, FLEXIBLE, TECHNICAL MANUAL

This specification is approved for use by the Department of the Air Force, and is available for use by all Departments and Agencies of Defense.

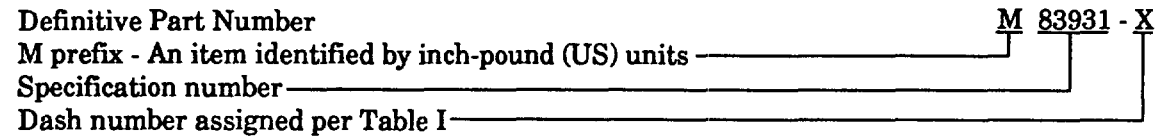
## 1. SCOPE.

1.1 This specification covers the requirements for three types of binders used to house technical manuals.

1.2 Classification. Binders covered by this specification are of the following types, as specified (see 3.2 and 6.2):

| TYPE | DESCRIPTION |
|---|---|
| I | Standard Size |
| II | Pocket Size |
| III | Pocket Size, Modified |

1.2.1 Definitive part number. The definitive part number is formatted to identify each item covered by this specification. The part number is formatted by selecting the options available in this specification as follows (see 3.4):




Example: M83931-1 is standard size binder for 8.25 x 10.75 manuals.

1.2.1.1 Classification designator. A one-position field used to designate the type of binder to the assigned dash number (see Table I).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Warner Robins Air Logistics Center, ATTN: MMIRFW, Robins AFB, GA 31098-5609 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A

FSC 7510

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TABLE I

| DASH NO. | TYPE | BINDER TYPE | MATERIAL ACCOMMODATION SIZE |
|---|---|---|---|
| -1 | I | Standard | 8.25 x 10.75 inches (210 x 273mm) |
| -2 | II | Pocket | 4.00 x 8.00 inches (102 x 203mm) |
| -3 | III | Pocket, Modified | 5.00 x 8.00 inches (127 x 203mm) |

2. APPLICABLE DOCUMENTS.

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified, the following specifications and standards of the issue currently listed in the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein (see 6.2).

SPECIFICATIONS

Federal

PP-B-636 Box, Fiberboard

STANDARDS

Federal

FED-STD-595 Colors

Military

MIL-STD-129 Marking for Storage and Shipment

(Copies of specifications and standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS.

3.1 Materials. Recycled and recovered raw materials should be used to the maximum extent possible in lieu of virgin raw materials, as long as these materials do not jeopardize the intended use and fully comply with all contract requirements. Material used shall be free from defects which would adversely affect the performance or maintainability of individual components or the overall assembly.

3.1.1 Type I covers and backing strip. The covers and backing strip shall be flexible Lexide, Pyroxylin coated, latex base, blue color No. 25053 from FED-STD-595, of thickness 0.025 inches (0.635mm) plus or minus 0.003 inches (0.076mm).

3.1.2 Type II and III covers. The covers shall be one-piece, flexible, simulated leather coated on rubber impregnated base, blue color No. 235053 from FED-STD-595, of a thickness 0.025 inches (0.635mm) plus or minus 0.003 inches (0.076mm).



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3.2 Design and construction.

3.2.1 Type I. Is a standard size flexible binder to accommodate 8.25 x 10.75 inch (210 x 273mm) technical manuals and shall be constructed as shown in figures 1 and 2, and as follows:

3.2.1.1 Configuration. The binder shall consist of a front and back cover (figure 1) and backing strip (figure 2) with scored hinges. Manuals will be held in the binders by three metal screw posts inserted through the front and back covers, the appropriate holes in the backing strip, and the manual.

3.2.1.2 Size. Both front and back covers shall be the same size, 9.375 x 11.000 inches (238 x 279mm), plus minus 0.031 inches (0.79mm) in either dimension. The corners opposite the binding edge shall be rounded to a 0.375 inch radius. The backing strip shall be 4.25 x 11.00 inches (108 x 279mm), plus or minus 0.031 inches (0.79mm) in either direction. The backing strip shall be satisfactory for accommodating manuals varying in thickness from 0.050 to 3.50 inch (1.27 to 89mm), plus 0.500 inch (12.7mm) or minus 0.125 inch (3.2mm).

3.2.1.3 Screw post holes. Both covers shall have three 0.250 inch (6.4mm) diameter holes, 3.50 inches (8.9mm) center to center, 2.00 inches (50.8mm) from each end, 0.312 inches (7.9mm) from center of all holes to outer edge of hinge (see figure 1). The backing strip shall be holed and scored per figure 2.

3.2.2 Type II. Is a six-ring binder to accommodate 4.00 x 8.00 inch (102 x 203mm) manuals or cards, and shall be as illustrated by figures 3, 4, and 7.

3.2.2.1 Configuration. The binder shall consist of a one-piece cover material and a loose-leaf six-ring base, attached and centered in such a manner that the binder is one single unit.

3.2.2.2 Cover and backing strip. The cover material shall be of one-piece flexible, simulated leather coated on rubber impregnated base, blue color No. 25053 of FED-STD-595, thickness of 0.025 inches (0.635mm), plus or minus 0.003 inches (0.076mm).

3.2.2.3 Ring Mechanism. Each ring mechanism shall be metal constructed, commercially available 1.00 inch (25.4mm) diameter rings, having a 0.625 inch (1.6mm) opening separation. The opening mechanism shall be a component part of the enclosed base and not protrude outside of the binder at any time after assembly.

3.2.2.4 Size. The size of the front and back cover shall be sufficient to prevent damage to enclosed material when the binder is laid or stood in any fashion and closed, but not to exceed a width of 12.00 inches (304mm) when laid open.

3.2.3 Type III. Is a six-ring binder to accommodate 5.00 x 8.00 inch (127 x 203mm) manuals or cards and shall be as illustrated in figures 5, 6, and 7.
3.2.3.1 Configuration. The binder shall consist of a one-piece cover material and loose-leaf, six-ring base attached and centered in such a manner that the binder is one single unit.

3.2.3.2 Cover and backing strip. The cover material shall be one-piece, flexible, simulated leather coated on rubber impregnated base, blue color No. 25053 of FED-STD-595, thickness of 0.025 inches (0.635mm), plus or minus 0.003 inches (0.076mm).



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3.2.3.3 Ring mechanism. Each ring mechanism shall be metal constructed and commercially available 1.500 inch (38mm) diameter with a 1.00 inch (25.4mm) opening separation. The opening mechanism shall be a component part of the enclosed base and not protrude outside the binder at any time after assembly.

3.2.3.4 Size. The size of the front and back cover shall be sufficient to prevent damage to the enclosed material when laid or stood in any fashion and closed, but not to exceed a width of 14.75 inches (375mm) when laid open.

3.3 Finish.

3.3.1 Cover printing. The front covers, on the outside surface, shall be imprinted with the words "USAF TECHNICAL MANUAL" in permanent white color, in the position shown in figure 1 for Type I and in figure 7 for Types II and III. The letters shall be capitals, 36 point type, Gothic style or similar.

3.3.2 Backing strips. The backing strips shall have white rectangular blocks, durable enough to withstand marking with a ball-point pen, and erasure of black lead pencil markings. These blocks shall be positioned as shown in figure 2 for Type I, figure 4 for Type II, and figure 6 for Type III.

3.3.3 Binder. The overall binder shall be free of decorations or decorative finishes.

3.4 Part identification. Each binder shall be marked with the specification part number (see 1.2.1), on the inside lower center of the front cover 1.00 inch (25.4mm) from the bottom.

3.5 Workmanship. The binders furnished under this specification shall be of commercial quality and free of cuts, scratches, breaks, tears, holes, sharp edges, or any other defect which would affect their serviceability. All title blocks and printing shall be clear and have sharp contrast, without smeared or smudged edges.

4. QUALITY ASSURANCE PROVISIONS.

4.1 Responsibility for inspections. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities, or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specifications where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Sampling.

4.2.1 Inspection lot. All binders of the same type offered for delivery at one time shall be considered a lot for purposes of sampling.

4.2.2 Visual and dimensional inspection sample. From each inspection lot, a random sample of binders shall be taken for Government inspection of visual and dimensional characteristics in accordance with MIL-STD-105.



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TABLE II - CLASSIFICATION OF DEFECTS - ALL TYPES

| Defect | Category Major | Minor |
|---|---|---|
| Dimensions of component parts not as specified. | X | |
| Binder does not satisfactorily accommodate referenced size manuals. | X | |
| Thickness of cover material different than specified. | X | |
| Cover not of the material specified. | X | |
| Cover not the color specified. | | X |
| Title block on back missing. | X | |
| Title block not specified size. | | X |
| Title block not durable enough to withstand marking and erasure as required by 3.3.2. | X | |
| Location, color, design, or durability of letting not as specified. | | X |
| Workmanship not as required by 3.5. | | |
| TYPE I | | |
| Scored hinges missing. | X | |
| Dimensions or spacing of holes in covers or backing strip not as specified. | X | |
| TYPES II and III | | |
| Ring mechanism not size specified. | X | |
| Opening boosters protrude beyond edge of cover. | | X |
| Opening boosters interfere with pages/manuals in binder. | X | |
| Opening boosters will not open rings. | X | |
| With rings closed, the mechanism does not lock rings together. | X | |
| Ring mechanism not securely attached to cover. | X | |



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4.3 Visual and dimensional inspection. Visual and dimensional inspection of each sample taken per 4.2.2 shall be in accordance with the classification of defects specified in Table II. The Acceptable Quality Level (AQL) shall be 1.5 percent defective for major defect and 2.5 percent defective for minor defects, and the inspection shall be Level II in accordance with MIL-STD-105. The Government reserves the right to classify as a defect any departure from the specified not specifically listed in Table II.

4.4 Preparation for delivery inspection. Packing and marking shall be inspected to determine conformance to the requirements of Section 5.

5. PREPARATION FOR DELIVERY.

5.1 Packing. Packing shall be Level A, B, or C as specified (see 6.2).

5.1.1 Level A. Binders shall be packed directly into PPP-B-636, Box, Fiberboard, Style RSC, Type CF, Class Weather Resistant. Containers shall be closed and contents waterproofed by taping all seams, corners, and manufacturer's joints in compliance with the box specification appendix.

5.1.2 Level B. Same as Level A, except waterproofing of contents shall not be required.

5.1.3 Level C. Binders shall be packed in a manner which will afford adequate protection against damage during domestic shipment from the supply source to the first receiving activity for immediate use. This level shall conform to the applicable carrier's rules and regulations, and may be the supplier's commercial practice, provided the latter meets the requirements of this level.

5.2 Marking. In addition to any special marking required by the contract or order, unit packages, intermediate packages, and shipping containers shall be marked in accordance with the requirements of MIL-STD-129.

6. NOTES.

6.1 Intended use. The flexible binders covered by this specification shall be used to hold standard size and pocket size technical manuals, including preliminary manuals.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number and date of this specification.

b. Specification part number and type of binder required (see 1.2).

c. Selection of applicable level of preservation, packaging, packing and marking required (see Section 5).



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6.3 Supersession. This specification supersedes that portion of MIL-M-4401E(USAF), 1 May 1967, that pertains to binders.

6.4 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the change.

Custodians:
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Preparing activity:
Air Force - 84

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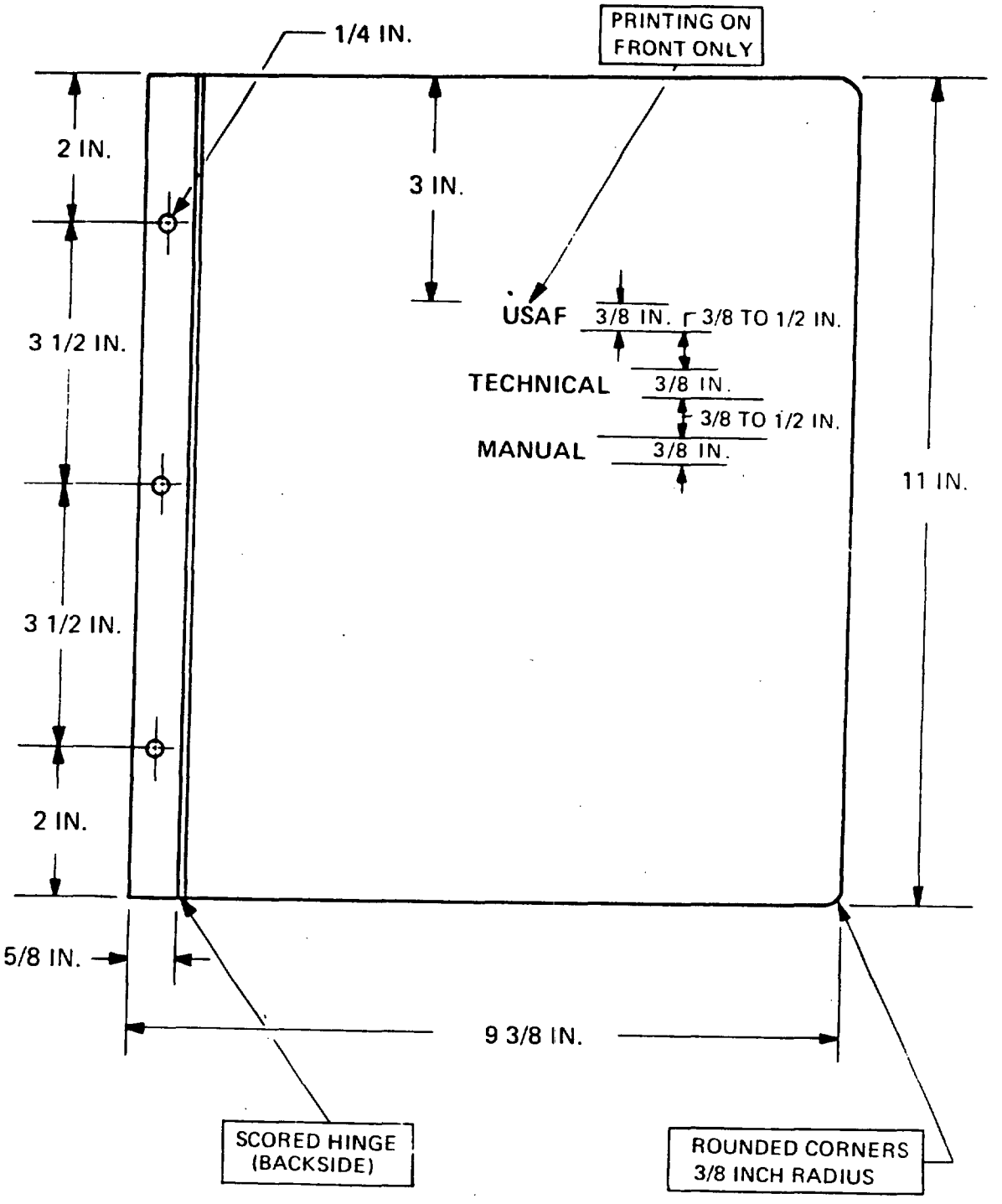



Figure 1. Type I Binder Cover Dimensions



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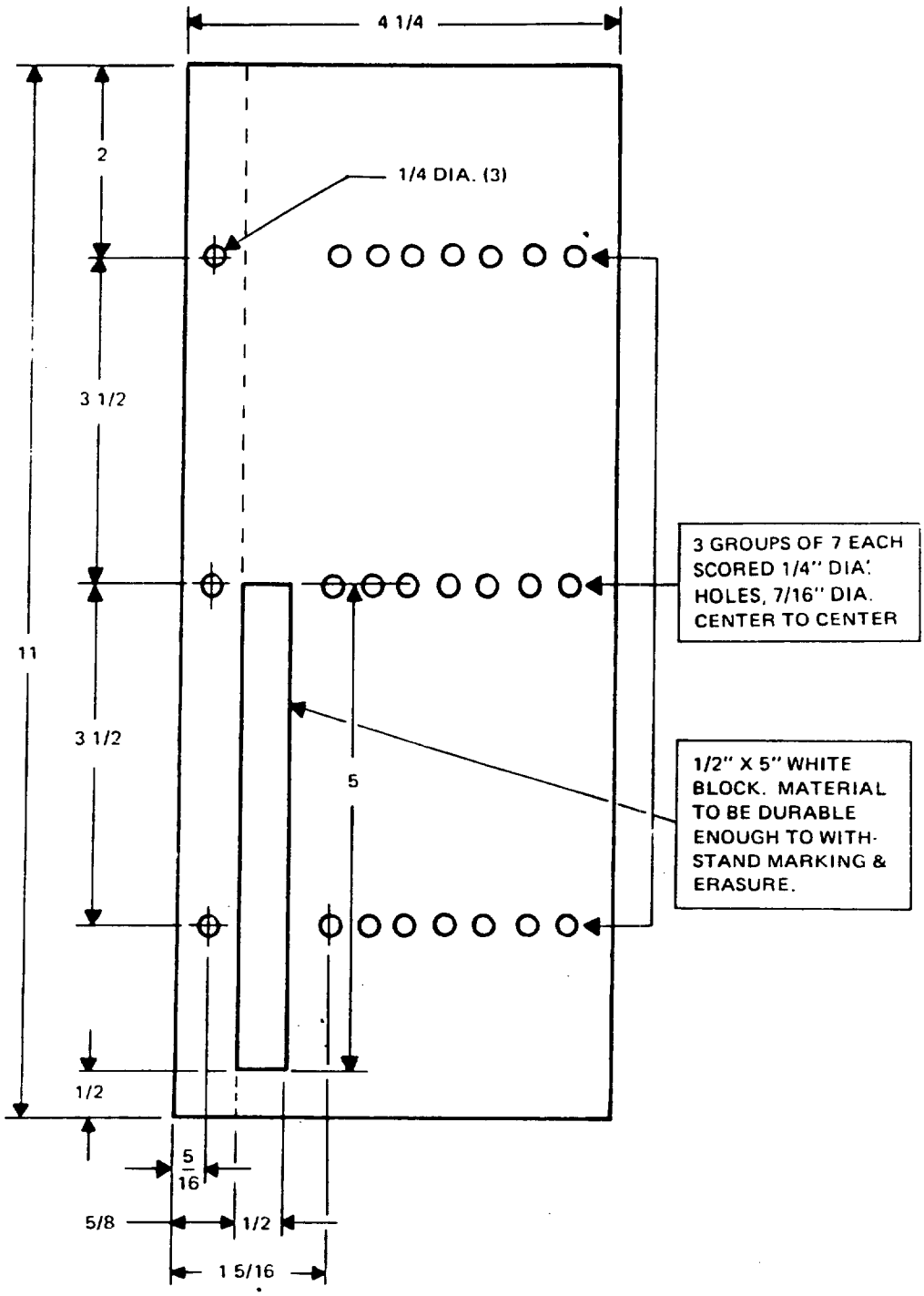



Figure 2. Type I Binder Backing Strip Dimensions



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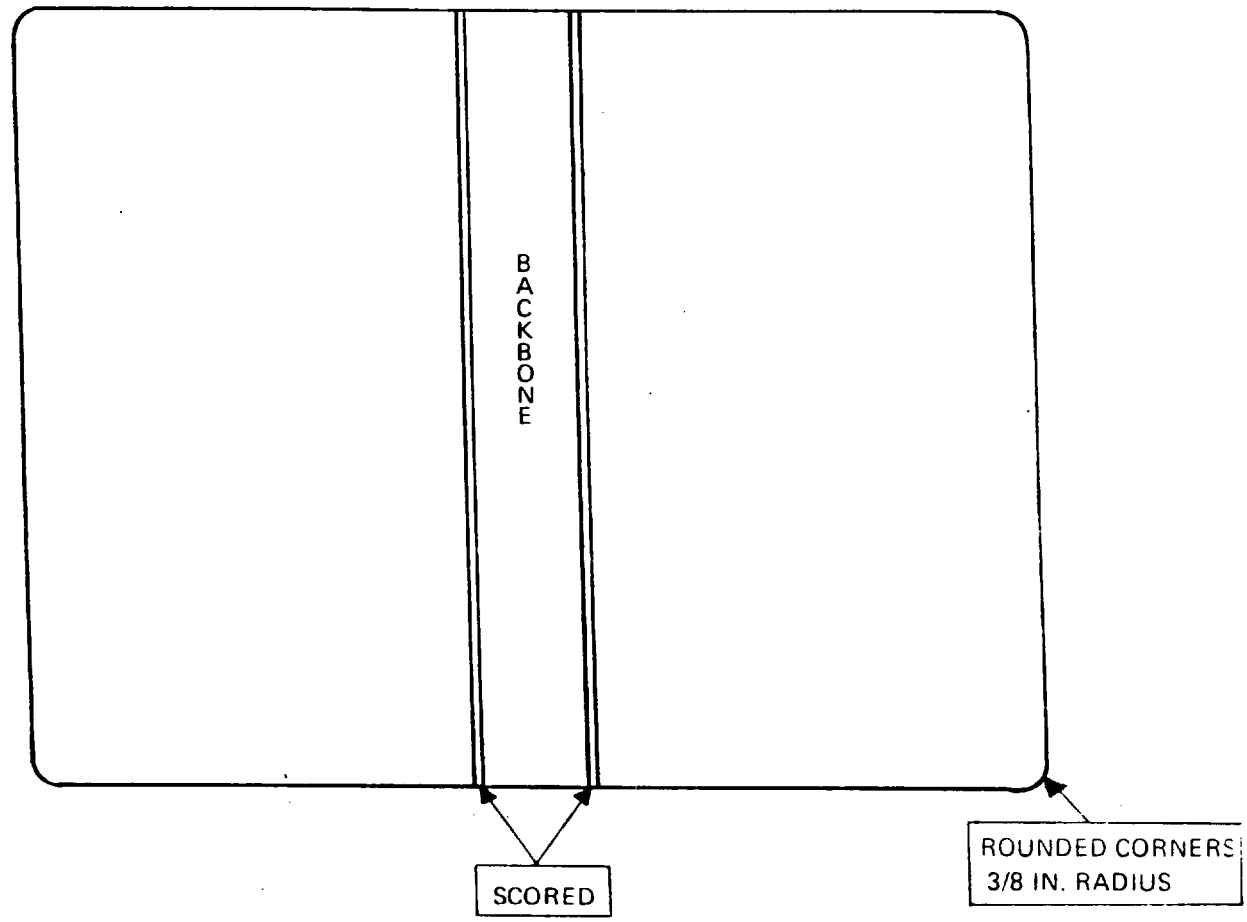



Figure 3. Type II Binder Cover Dimensions, Pocket Size (Inside)



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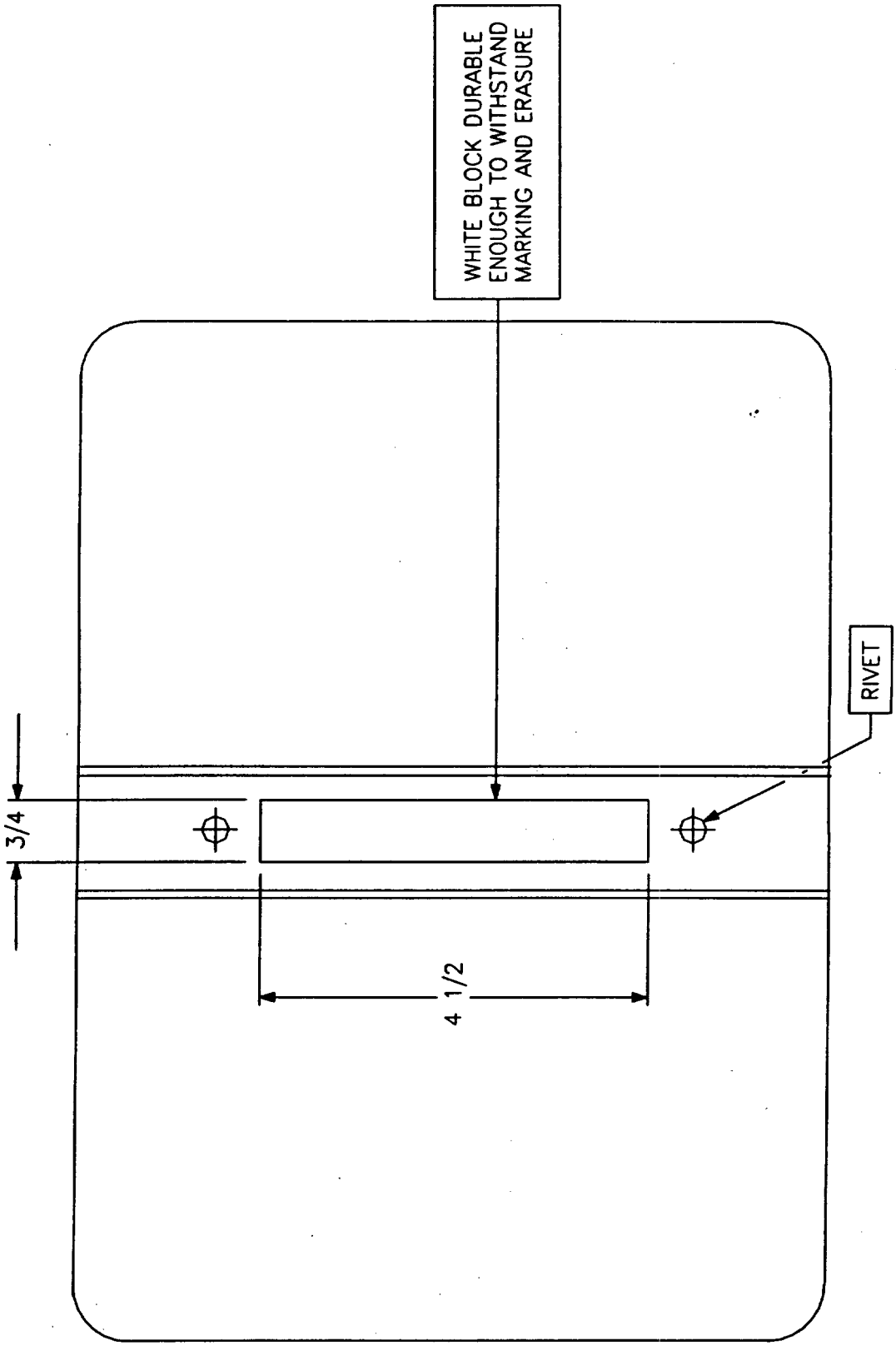



Figure 4. Type II Binder Backing Strip Dimensions. Pocket Size (Outside)



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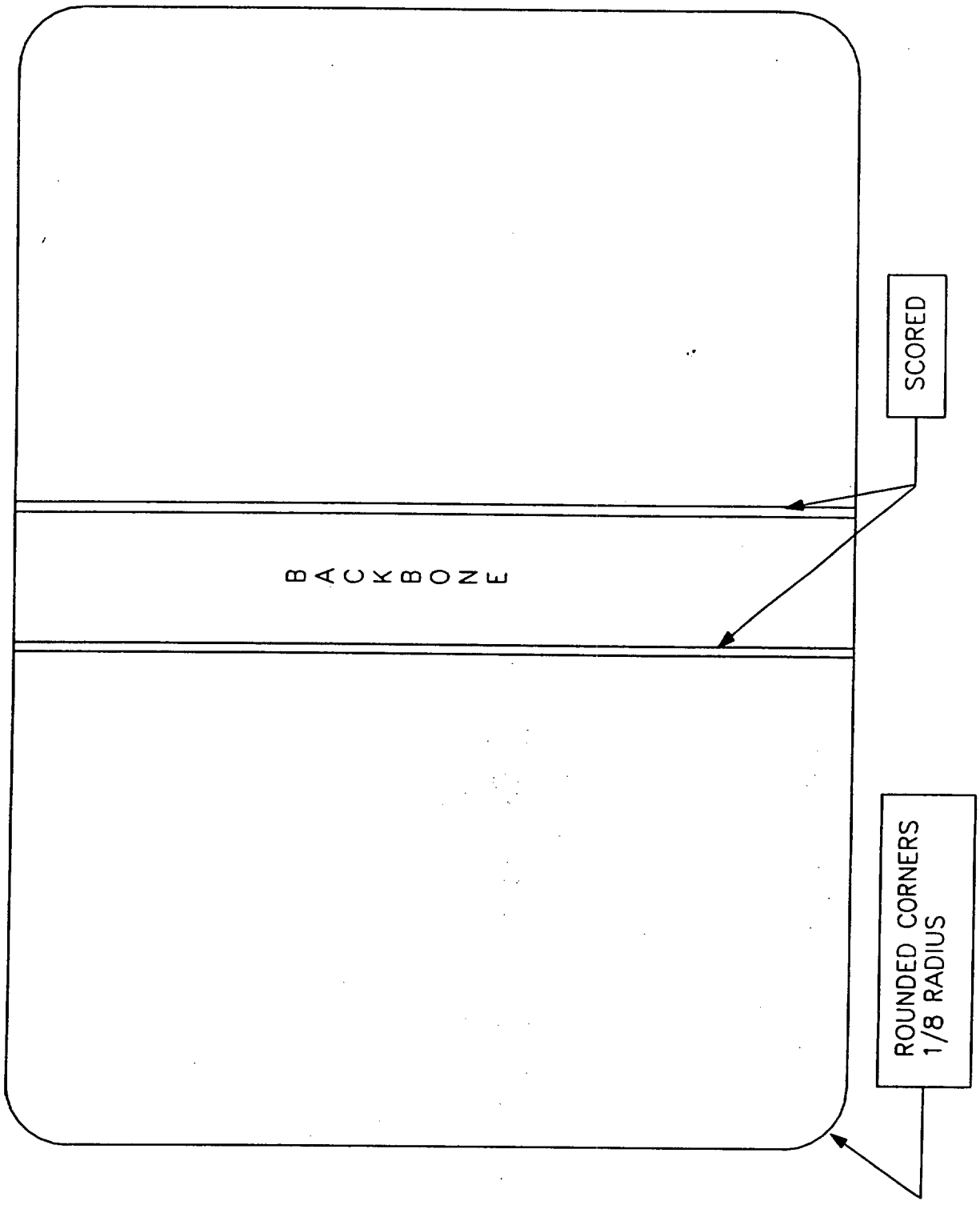



Figure 5. Type III Binder Cover Dimensions (Inside)



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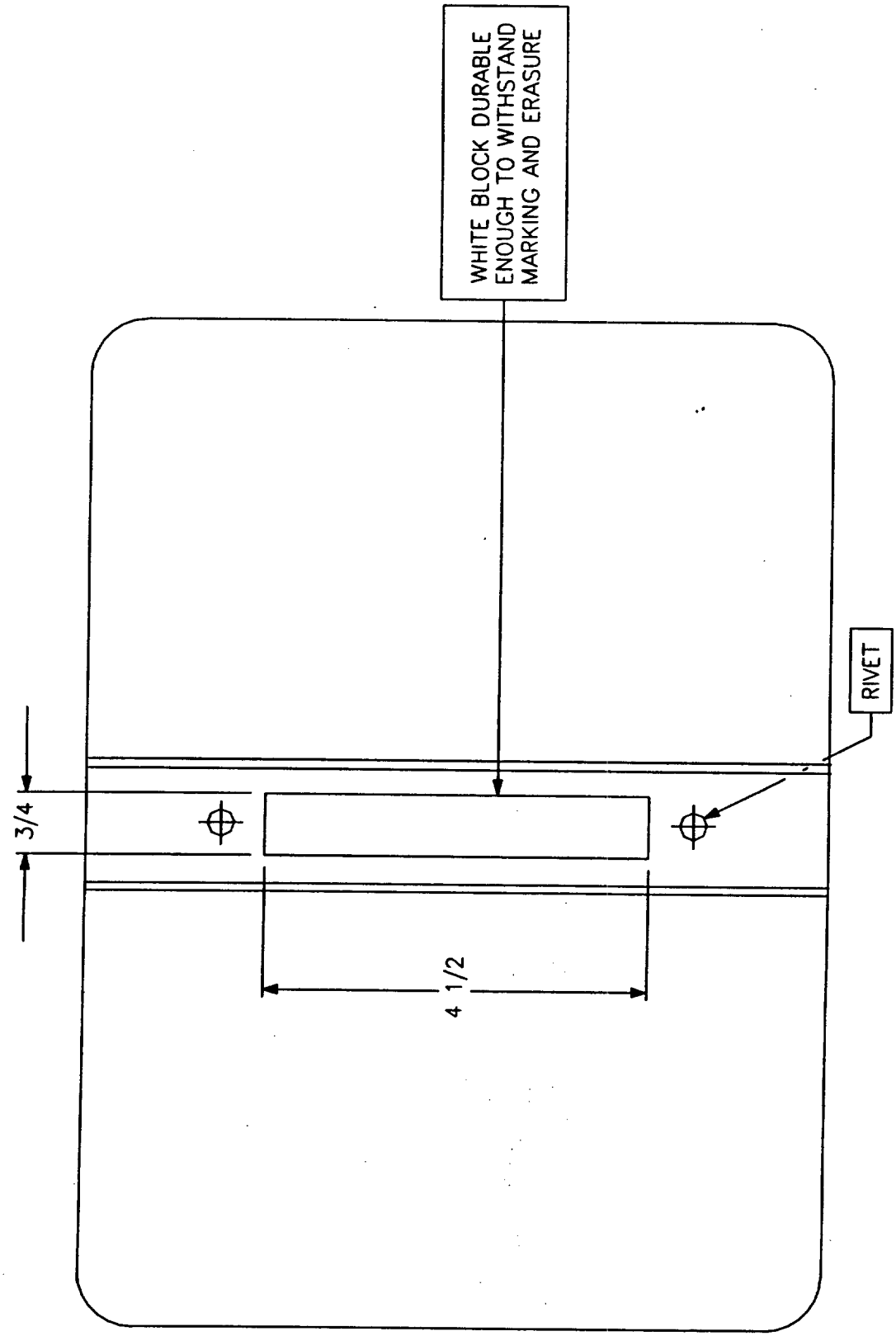



Figure 6. Type II Binder White Block Dimensions (Outside)



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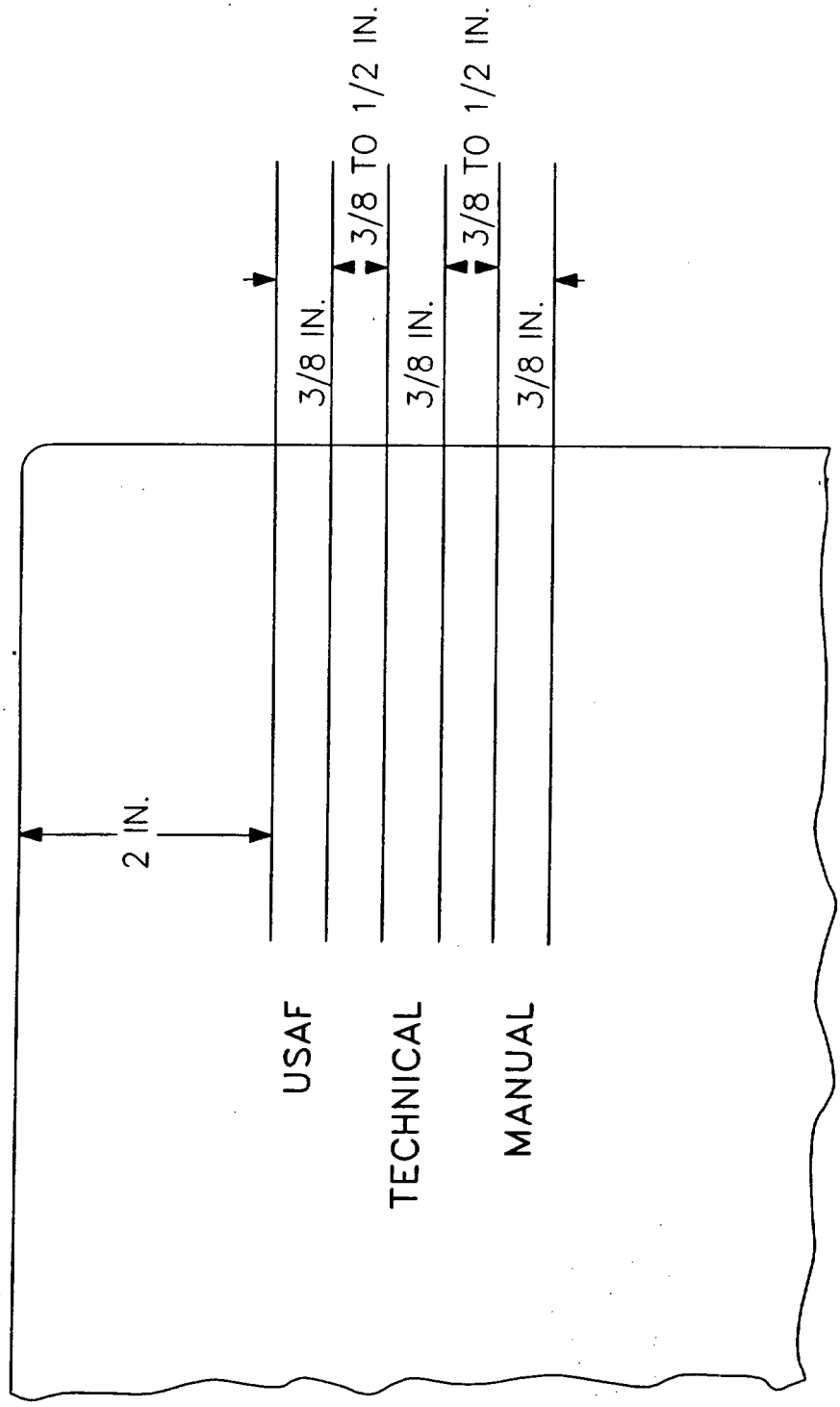



Figure 7. Lettering on Types II and III Binders
(Outside right sideview)

Note: Centrally locate in horizontal direction and comply with above dimensions.



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